BEYOND WORD

TEAM NAME : VISIONX

TEAM MEMBERS: HEMASRI

: VISMAYA

TABLE OF CONTENTS:

1.Problem statement 01

2.System Architecture 02

3. Workflow 03

4.Technologies Used 04

5.Key Features and functionalities 05

6.Explanation of How solution works 06-09

1. Problem Statement

Most web content today is designed for information density rather than readability. Articles, blogs, and study materials are often presented as large blocks of text surrounded by advertisements, navigation panels, and other visual distractions.

While this format works for many readers, it creates significant challenges for individuals with learning differences such as **dyslexia and ADHD**. These users may struggle to track lines of text, maintain focus, or process written information efficiently. As a result, reading digital content can become tiring and frustrating.

Considering that a significant portion of the population experiences some form of neurodivergence affecting reading and attention, the current design of web content creates an unnecessary barrier to accessing information.

The goal of this project is to **improve the readability of digital content by adapting the presentation of text to support different cognitive reading styles**. Instead of expecting every user to adjust to dense webpage layouts, the system modifies the content dynamically to make reading clearer, more structured, and easier to follow.

2. System Architecture

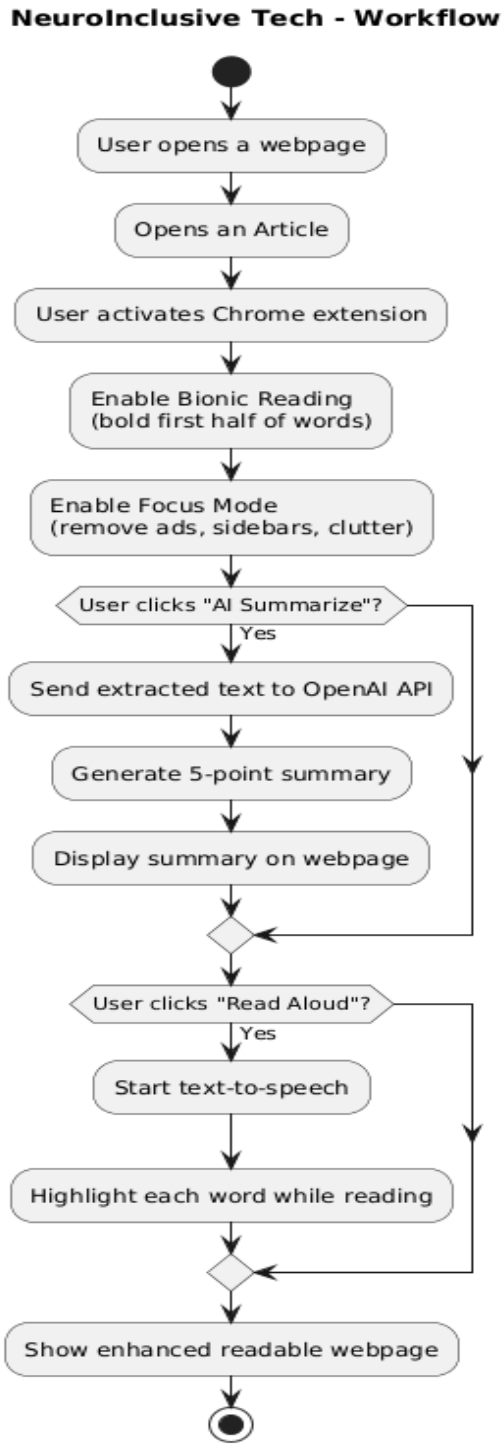
The system is implemented as a **Chrome browser extension** that operates directly on webpages visited by the user. The extension analyzes the page content and modifies its presentation to improve readability.

Architecture Overview

1. **User Layer**
   - The user opens any webpage through the browser.
2. **Chrome Extension Layer**
   - The extension activates automatically and provides controls for enabling different reading assistance features.
3. **Content Script**
   - The content script interacts with the webpage's Document Object Model (DOM) and extracts the main textual content.
4. **Processing Layer**
   - The extracted text is processed through different feature modules that modify the presentation of the content.
5. **AI Integration**
   - When the user requests a summary, the relevant text is sent to the OpenAI API for summarization.
6. **Output Layer**
   - The processed content is rendered back on the webpage, providing an improved reading experience.

This architecture allows the extension to operate **locally within the browser while selectively using AI services for summarization tasks**.

3.Workflow


NeuroInclusive Tech - Workflow
User opens a webpage
Opens an Article
User activates Chrome extension
Enable Bionic Reading
(bold first half of words)
Enable Focus Mode
(remove ads, sidebars, clutter)
User clicks "AI Summarize"?
Yes
Send extracted text to OpenAI API
Generate 5-point summary
Display summary on webpage
User clicks "Read Aloud"?
Yes
Start text-to-speech
Highlight each word while reading
Show enhanced readable webpage

4. Technologies Used

The project is built using technologies that support browser extensions and dynamic webpage modification.

Frontend Technologies

- HTML
- CSS
- JavaScript

Browser Extension Framework

- Chrome Extension API
- Content Scripts
- DOM Manipulation

Styling and Page Modification

- CSS Injection

Artificial Intelligence Integration

- Gemini ai API (for webpage summarization)

These technologies allow the system to dynamically modify webpage content while maintaining compatibility with most websites.

5. Key Features and Functionality

This extension provides several features designed to improve readability and reduce cognitive load for users.

Bionic Reading Mode

This feature modifies the text by highlighting the first portion of words. This visual emphasis helps guide the reader's eye through the sentence and improves reading flow.

Focus Mode

Focus Mode removes unnecessary visual elements such as advertisements, sidebars, and clutter from the webpage. The content is displayed in a clean and simplified layout that reduces distractions.

AI-Powered Summary

Users can generate a short summary of the webpage using the OpenAI API. The extension extracts the main text from the page and produces a concise set of key points that capture the essential information.

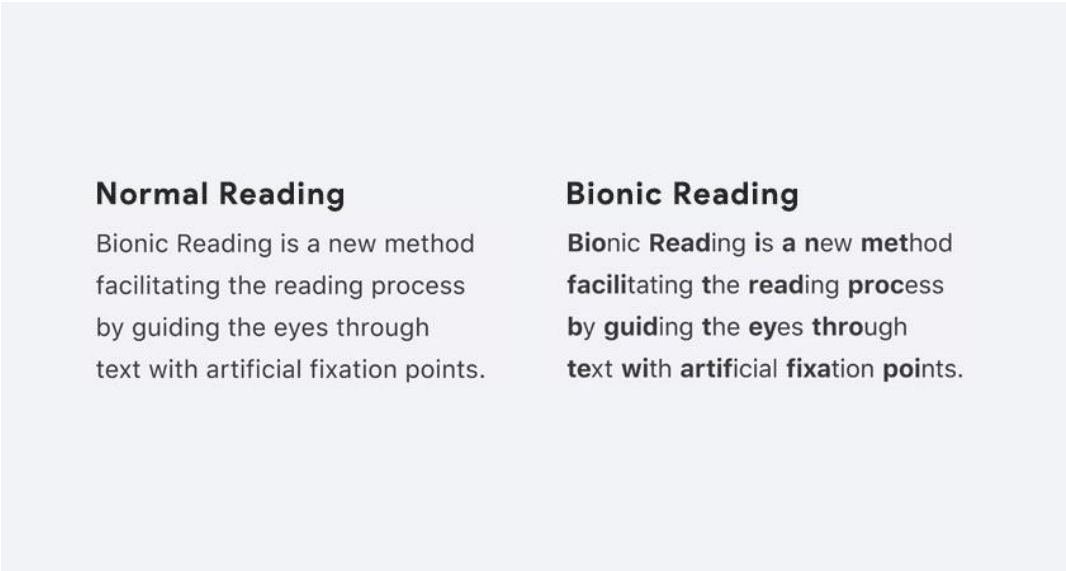
Read-Aloud with Word Highlighting

The extension includes a text-to-speech feature that reads the webpage content aloud. As the text is spoken, each word is highlighted on the screen, allowing users to follow the reading visually while listening.

6. Explanation of How the Solution Works

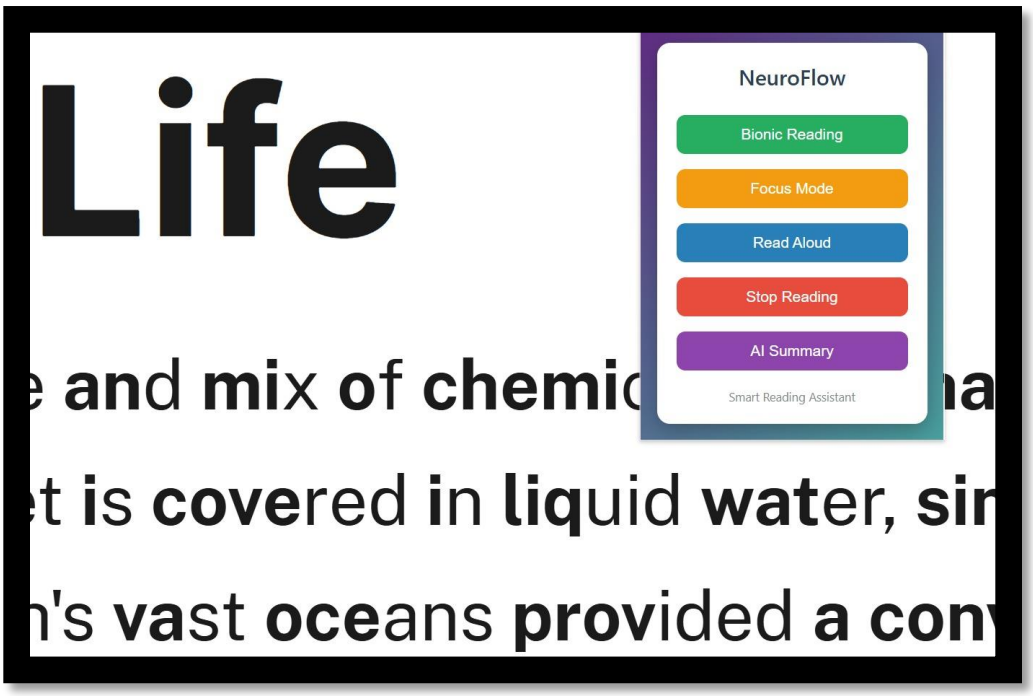
The solution works by intercepting and modifying webpage content directly within the browser.

When a user opens a webpage, the Chrome extension automatically activates and injects a content script into the page. This script extracts the main textual content from the Document Object Model.

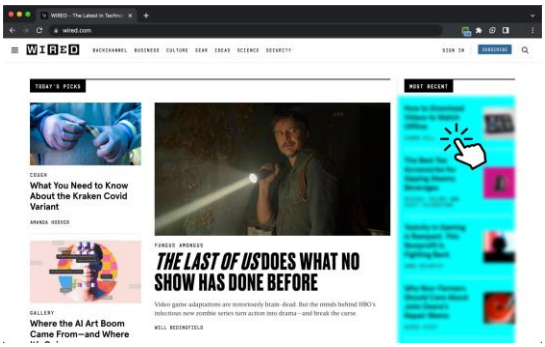


Once the text is extracted, the extension applies readability enhancements through different processing modules. The Bionic Reading module modifies word formatting to improve visual guidance, while the Focus Mode module simplifies the layout by hiding non-essential elements.

BIONIC TEXT

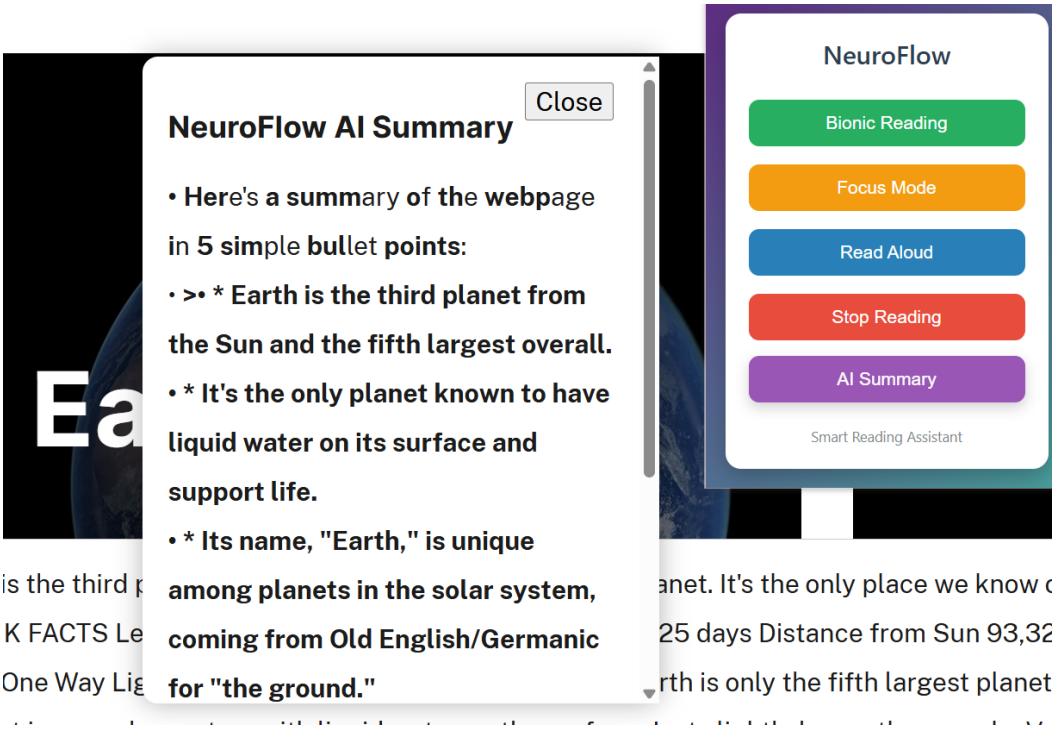


BEFORE FOCUS MODE ON



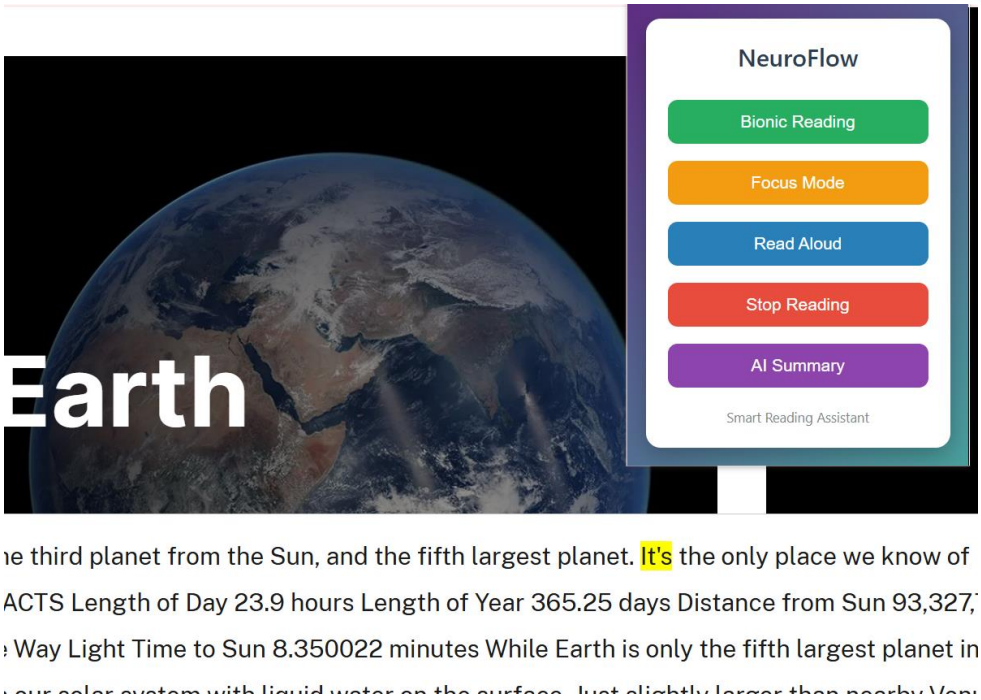
AFTER FOCUS MODE ON

If the user chooses to generate a summary, the extracted text is sent to the OpenAI API. The response from the API is processed and displayed as a concise summary on the page.



AI SUMMARIZER IN 5 BULLET POINTS

The Read-Aloud feature uses browser-based text-to-speech capabilities to read the content aloud while synchronizing visual highlighting with the spoken words.



HIGHLIGHTING THE WORD WHIE READING ALOUD

Through these combined features, the extension transforms standard webpages into a **more accessible and readable format**, enabling users with different reading styles to engage with digital content more effectively.